

Python Data Types Theory Questions Quiz

1. What is the difference between a list and a tuple in Python?

- A) Lists are mutable, tuples are immutable
- B) Lists are immutable, tuples are mutable
- C) Both are mutable
- D) Both are immutable

Correct Answer: A

Explanation: Lists can be changed (mutable), while tuples cannot be changed after creation (immutable).

2. Explain the concept of mutability with respect to Python data types.

- A) It determines whether an object can be changed after creation
- B) It shows how fast an object is
- C) It defines the data type's size
- D) It affects only numbers

Correct Answer: A

Explanation: Mutability means whether an object's value can be changed after it is created.

3. What happens when you assign one list to another variable and then modify the original list?

- A) The second list remains unchanged
- B) Both variables reference the same list, so both show changes
- C) It creates a new list
- D) An error occurs

Correct Answer: B

Explanation: Python assigns by reference, so both variables point to the same list.

4. How is a tuple different from a list in terms of usage and performance?

- A) Lists are faster and used for fixed data
- B) Tuples are faster and used for fixed data
- C) Lists consume less memory
- D) Tuples are mutable

Correct Answer: B

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Explanation: Tuples are immutable, so they're optimized for performance and memory usage.

5. Why would you use a tuple instead of a list?

- A) When you need an immutable sequence
- B) To store only numbers
- C) For changing data
- D) For larger memory usage

Correct Answer: A

Explanation: Tuples are used when data must not change, like keys in a dictionary.

6. Describe how dictionaries store data in key-value pairs.

- A) By using a list structure
- B) Using a hash table implementation
- C) Through ordered pairs only
- D) With tuple sets

Correct Answer: B

Explanation: Dictionaries use hash tables for efficient storage and lookup.

7. Can a dictionary have duplicate keys?

- A) Yes, duplicate keys are allowed
- B) No, each key must be unique
- C) Only integers can be duplicate
- D) Only strings can be duplicate

Correct Answer: B

Explanation: Keys in dictionaries must be unique to avoid confusion in lookup.

8. How can you update the value of a key in a dictionary?

- A) By appending to the key
- B) By assigning a new value to the key
- C) Using the insert() method
- D) Using add() method

Correct Answer: B

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Explanation: Simply assign a new value using `dict[key] = new\_value`.

9. What will happen if you try to access a key that doesn't exist in a dictionary?

- A) It returns 0
- B) It returns None
- C) It raises a KeyError
- D) It adds the key automatically

Correct Answer: C

Explanation: Accessing a missing key without using `.get()` will raise a `KeyError`.

10. What are some common methods used with dictionaries?

- A) push(), pop(), add(), remove()
- B) keys(), values(), items(), get()
- C) map(), filter(), reduce(), zip()
- D) insert(), delete(), update(), select()

Correct Answer: B

Explanation: These methods help in safely accessing and managing dictionary data.